# 1 Pharmacokinetic Data Structures

The CDISC(Clinical Data Interchange Standards Consortium) data submission models contain a domain for pharmacokinetic data. The CDISC model includes structures for the transfer of PK Concentration Data and PK Parameter Data. As such, the PKOne output data file should conform to that model. Dataset structures have gone through internal ABC review CDISC and IDSL compliance. The pk datasets consist of the following:

- PK: Pharmacokinetic Sample Data (from the CRF)
- PKCNC: Pharmacokinetic Concentration Data (post merge with the pk concentration data)
- PKPAR: Pharmacokinetic Parameter Data

## 1.1 Rationale:

The CDISC data structures target the datasets that are submitted to the FDA and it seems reasonable to have the intermediary data-sets comply with the final standard as much as possible. This will prevent tweaking files during submission time when pressure on delivery is high, and ensures that all of the data that are required in the CDISC model will be collected and stored in a central location.

## 1.2 Principles:

PK parameters will be denoted as described in the list of "Standard Parameter Nomenclature for Reports Containing Pharmacokinetic Data". This list contains the most parameters that are derived for analysis. As other parameters become available in the data, a standard must be defined in order to implement an automated system.

- PK parameter names to be used in the text of reports will match the format listed in the first column of the standard nomenclature list (Report Text Symbol(s)).
- PK parameter names to be used in SAS output will match the format listed in the second column of the standard nomenclature list (SAS report format).
- PK parameter names that are created by WinNonLin contain special characters that are not recognized by SAS. Thus, the WinNonLin parameter names will be converted to the names listed in the third column of the standard nomenclature list to facilitate the transfer of the data into SAS.

## 1.3 Rationale:

The development of a standard format for PK parameter nomenclature will provide consistency between study reports, and allow for modular programming.

# 2 Pharmacokinetic Table Display Principles

Please refer to SOPs :

- Standard Methods for the Non-Compartmental Analysis of Pharmacokinetic Data (SOP/CPK/0001/002).

- Standard Statistical Methods for the Analysis of Pharmacokinetic Data (SOP/BMD/4002/v01)

Also please reference Statistical Display Principles 5.01-5.08.

Of particular interest:

- Principle 6.06.1: Summary Tables - Minimum Set of Statistics for Numeric Variables:

  The minimum set of summary statistics for numeric variables are: n, mean, standard deviation (or standard error), median, minimum, and maximum. This principle does not apply to the output of statistical models or when the standard list is inappropriate.

- Principle 6.06.2: Summary Tables - Choice of Variability Statistic:

  If variables such as baseline characteristics are being summarised for the purpose of comparing distributions of treatment groups, then standard deviations should be displayed. For the purposes of comparing estimates (e.g., treatment difference in blood pressure), standard errors should be displayed. Standard deviations describe the distribution of the data, while standard errors describe the precision of an estimate.

%CVb (CPSP Specific)

There are two methods used for the calculation of the between subject coefficient of variation as stated in SOP- BMD-4002:

1. %CVb=100*(SD/Mean) with SD and Mean of untransformed data
2. %CVb=100*sqrt($e^{SD**2}$-1) with SD of log-transformed data

Method 1 will be used when %CVb is reported on Table 3 and Method 2 will be used when %CVb is reported in Table 2.

- Principle 6.06.3 Summary Tables - Precision for Numeric Variables: The minimum and maximum values are presented with the same number of decimal places as the raw data collected on the CRF. The mean and percentiles (e.g. median, Q1, and Q3) are presented using one additional decimal place. The standard deviation and standard error are presented using two additional decimal places. Within a column, summary statistics are aligned along the decimal point.

- Principle 6.10 Summary Tables - Presentation of Confidence Intervals Confidence intervals are displayed within round brackets and separated by a comma. The lower limit is presented first, and the upper limit is presented second. The confidence interval is placed either beside or below the corresponding estimate. The estimate and the confidence limits are presented using the same degree of precision. The label or description of the confidence interval includes the percentage range (e.g., 95%). The standard abbreviation for confidence interval is CI.

## 2.1 Precision of PK Concentration Data

Raw PK concentration data is released by the bioanalytical laboratory. The lower limit of quantification (LLQ) of the PK concentration values will be provided.

As a general principle for PK concentration data released by the bioanalytical laboratory via SMS2000 will be reported to a fixed number of decimal places as 3 significant figures at the LLQ.

As a general principle for PK concentration data released by the bioanalytical laboratory via Watson LIMS will be reported to 3 significant figures.

If the PK concentration data have more decimal places than the LLQ value, then imputed PK concentration data may be rounded to the number of decimal places of the LLQ value (the LLQ will normally be provided to a precision of 3 significant figures.) For example, if the LLQ value is 5.00 and the PK concentration data have more than two decimal places, then the PK concentration data may be rounded to two decimal places.

## 2.2 Precision of PK Parameters from WinNonlin

Derived PK parameters, the numeric values output from the WinNonlin application do not constitute "raw data" as they are derived from other values. PK parameters generated by WinNonlin are not rounded prior to delivery for statistical analysis, and often contain many significant figures. As such it is appropriate to define a sensible accuracy to which these values should be reported. PK parameter values will generally be reported to a number of decimal places consistent with 3 significant figures for the lowest value for that parameter, unless the original data for that parameter is reported to less precision throughout (in which case the precision of the original data will be used). This approach ensures that all values are reported to at least 3 significant figures, which is sufficient for all the commonly used parameters.

## 2.3 Special Non-numeric PK Concentration values

Non-numeric values may appear in the concentration data. These values include: NR = 'No Result', NA = 'Not Assayed', NS = 'No Sample', IS = 'Insufficient Sample', and NQ = 'Non Quantifiable'.

The standards for imputation of non-numeric PK concentration values are provided in Pharmacokinetic (PK) Concentration Analysis and Reporting Process Guidance [3] for both blood (plasma or serum) and urine data. Other sample types should be discussed with the pharmacokineticist.

## 2.4 Special Non-numeric PK Parameter values

Non-numeric values may appear in the parameter data. These values include:

Not determined (ND) meaning the PK parameter could not be adequately defined for reasons other than NQs. This may include (but not be limited to) the following:

- t1/2 – terminal elimination phase not adequately defined
- AUC(0-∞) extrapolation > acceptable limit
- CL not relevant for route
- AUC(0-inf) not relevant for repeat dose

The standard for imputation of ND is NULL.

Non Calculable (NC) meaning that the PK parameter could not be derived due to the presence of NQs.

The standard for imputation of NC is NULL.
The alternate for imputation is as for standard, except for the following parameters, where Cmax and Ctau are imputed as ½ LLQ and AUC is imputed as ½ lowest observed AUC within the dataset (for each analyte) across all treatments and days.

Additional guidance for imputation of NC values can be found in “Standards for the Handling of NQ impacted PK Parameters” [4].